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Cross Sections for Low-Energy (1–100 eV) Electron Elastic and Inelastic Scattering in Amorphous Ice

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We report the integral cross sections per scatterer (i.e. elastic collision, phonon excitations, vibrational excitations, electronic excitations and ionization) for 1–100 eV electron scattering in an amorphous film of ice condensed at a temperature of 14 K. The integral cross sections are determined relative to the total from a two-stream multiple-scattering analysis of the electron energy distribution backscattered from the film. Their energy dependence is obtained from both the analysis of the elastic electron reflectivity as a function of the film thickness and the vibrational electron energy-loss spectra measured for several incident energies and large film thickness. The magnitude and various features found in the energy dependence of the cross sections are discussed, whenever possible, by comparison with data and with scattering mechanisms available in the gas phase. Microscopic effects, which are implicitly included in cross sections determined in this way, are discussed in terms of interference and coherent multiple-scattering contributions among the scattering sites as well as interactions of the scattering sites with their neighbors in the condensed phase. © 2003 by Radiation Research Society

INTRODUCTION

Ionizing radiations, such as γ rays and charged particles at high speed, produce along their tracks a considerable number of electrons having kinetic energies below 100 eV (*1*). This stems essentially from the fact that cross sections (i.e. the generalized oscillator strengths) for ionization as well as electronic excitations of matter by particles at high speed are more important for small momentum and energy transfers. Here, one must consider the term “high speed” as being a velocity much greater than that of excited or ionized atomic or molecular electrons. In the case of a valence-shell electron, it is about the Bohr velocity, which corresponds to a kinetic energy of one Rydberg (i.e. 13.6

eV) (*1*). It is further estimated that as much as 10–20% of the initial energy of the ionizing radiation eventually degrades into kinetic energy of quasifree electrons in the sub-excitation region (i.e. below the ionization threshold) (*2, 3*). Owing to their sizable scattering cross sections, specificity in their energy transfer, and their presence in large number in matter, electrons with kinetic energies below 100 eV (defined hereafter as low-energy electrons) have been considered to be a key factor for radiation energy deposition processes occurring over times of 10^{-11} – 10^{-15} s (*1*).

In the gas phase, the nature and the yield of all initial molecular species arising from the full slowing down of electrons can be deduced from the various cross-section data for electron scattering from the individual targets along with their use in a classical transport model and Monte Carlo methods (*1, 4*). Since the measurement of cross sections for low-energy electron scattering in condensed phases is not trivial to perform and analyze, such data are not generally available for studying the action of low-energy electrons, especially in materials of biological interest (*5*). So far in the case of condensed water, except in the sub-excitation region (*6, 7*), cross-section data used in calculations or simulations have mainly been inferred from gas-phase measurements or predicted from theoretical methods (*8–11*). Still, the presence of neighbors around a scattering site can modify the scattering cross sections significantly from those known in the gas phase. In principle, the latter modifications should preclude the use of gas-phase data as input values for any method of analysis of low-energy electron collision processes and their effects in condensed matter.

In a previous series of papers (*12, 13*), we obtained, from high-resolution elastic-electron reflectivity measurements within the 1–18 eV range, the total cross section per scatterer for the scattering of electrons in amorphous ice film condensed at a temperature of 14 K. With the knowledge of this total cross section, the integral cross section per scatterer for elastic, vibrational and electronic excitation processes could readily be determined from a two-stream multiple-scattering analysis (*14*) of the electron energy-loss spectra of thick ice films.

In the present paper, we revisit our previous data for

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amorphous ice films and extend the measurements to 100 eV. In view of the wider incident energy range explored, the present sets supersede those of refs. (12) and (13). The scattering cross sections for elastic and various inelastic processes are determined as before by applying the same multiple-scattering analysis. The magnitudes of the extracted cross sections in amorphous ice generally lie between 2% and 80% of their gas-phase counterparts. The difference between the present data and those reported in the gas phase is discussed in terms of two types of microscopic effects. The first arises from the quantum mechanical nature of the electron when its de Broglie wavelength compares, within an order of magnitude, to the closest distance separating the atomic or molecular sites. As a result, supplemental interference terms in addition to those originating from the individual scattering sites are needed for a microscopic description of the scattering probabilities (i.e. cross sections) in a condensed medium. The second effect comes into play when the interaction of an atomic or a molecular site with its neighbors can no longer be neglected. The resulting perturbation is responsible for the screening of the interaction of an electron with an atom or molecule in condensed phase. This screening has an immediate effect on the strength of the elastic and inelastic scattering that would originate from noninteracting sites (i.e. isolated targets).

EXPERIMENT

The electron-scattering experiments were performed with a high-resolution electron-energy-loss (HREEL) spectrometer described in detail previously (15). The spectrometer is housed in an ion-cryopumped UHV system (16) at a typical base pressure of 5×10^{-11} Torr. The incident electron beam is provided by a monochromator, which can be rotated from $\theta_0 = 14$ to 70° about the sample's normal, whereas the backscattered electrons are detected by an analyzer, which is fixed at $\theta_d = 45^\circ$ at the opposite azimuth. Double-zoom electron lenses at the exit of the monochromator and at the entrance of the analyzer allow adjustment of the focusing of the incident and analyzed electron beam over a wide energy range (e.g. 1–50 eV), while keeping a relatively low-pass energy (i.e. 1–2 eV) within the deflectors. The present data were recorded with the monochromator oriented near to the normal of the sample at 14° . The combined resolution of the spectrometers was adjusted at 15–20 meV full width at half maximum (FWHM) for a current at the sample of about 0.5 nA. The incident electron energy, E_0 , was calibrated with respect to the vacuum level, within ± 0.1 eV, by measuring the threshold of the electron current transmitted through the sample. The latter measurement was also performed during an experiment to ensure that the sample did not charge due to the trapping of electrons or creation of holes. Furthermore, any sample decomposition and other effects could be detected by comparing spectra recorded at different times.

The samples initially prepared in a gas-handling manifold are condensed from gas or vapor phases onto a polycrystalline platinum substrate. The latter is isolated electrically and press-fitted with a ceramic plate to the tip of a closed-cycle cryostat reaching a temperature of 14 K. The manifold consists of two gas or vapor sources connected through bypass and precision-leak valves to a small calibrated volume. This volume is connected by an admission valve to a capillary having an opening located in front of the substrate. The latter was cleaned by resistive heating in UHV to a temperature of 1800 K and in the presence of oxygen to 1200 K. These treatments gave energy-loss spectra that were free from vibrational energy losses due to contaminating atoms or molecules (17).

They further allowed the observation of low-energy electron diffraction (LEED) threshold interference structures that are found in the energy dependence of the intensity of the electron beam reflected specularly from a clean platinum surface (18). The water sample consisted of triply distilled water degassed by several freeze-thaw cycles under vacuum. The purity of water was verified further *in situ* with a quadrupole mass spectrometer (0–200 amu) facing the target. The absolute molecular coverage (i.e. molecules cm^{-2}) onto the platinum substrate was determined to an accuracy of $\pm 15\%$ from the calibrated differential pressure drop within the manifold as described in detail previously (12). For convenience, the molecular coverage can be converted into a number of condensed layers by normalizing to the density of 5.72×10^{14} molecules cm^{-2} of the (111) face of the cubic form of ice at a temperature of 143 K (19).

Under conditions of small deposition rates (i.e. $\sim 4 \times 10^{13}$ molecules $\text{cm}^{-2} \text{ s}^{-1}$) and low temperature (i.e. 14 K), such as used in the present experiment, water is known to condense into the low-density amorphous form of ice (20). It was shown recently (21) that amorphous films of ice can be grown in a nonporous to a highly porous form, depending on the angle of incidence of the water beam with respect to the substrate surface and the condensation temperature. Multilayer film growth on a Pt(111) crystal at a temperature of 22 K was found to be relatively dense and smooth for near normal beam incidences but highly porous for large angle incidences. The “induction” coverage for developing the pore structure was determined to be typically 30–80 layers (22). For thicker films, the amount of pores increased linearly with the coverage, thus indicating a constant porosity per unit of thickness. In the present study, we believe that the growth of a water film could vary for the first few layers (i.e. < 10 layers) owing to the proximity of the platinum substrate. For the subsequent layers, which here concerns relatively small water coverage (i.e. 10 to 30 layers), the above considerations suggest that the film growths should be typical of the low-density amorphous ice that would admit some degree of porosity. However, the sample porosity as well as density somehow ends up to be transparent to the present analysis since the *cross sections* are to be determined essentially from the variation of the reflectivity with the *absolute molecular coverage*. Besides, variations in the thickness of a film are not expected to be too serious as long as they are smaller than the electron mean free path.

MULTIPLE-SCATTERING MODEL

The fate of electrons in amorphous solids can be described by the classical transport (i.e. Boltzmann) equation that includes stochastic elements of probability per unit length or scattering cross sections for momentum and energy transfers pertaining to various excitations of the medium (3). In this approach, the cross section is a quantity involving one or more atomic or molecular sites coherently scattering electrons within an effective volume and beyond which electron interference and correlation effects (i.e. between these volumes) are negligible. Within the first Born approximation, this quantity is proportional to the product of an atomic or molecular factor multiplied by the dynamic structure factor. In the absence of any correlation between scattering sites, such as in the gas phase, this definition reduces merely to the individual atomic or molecular cross section multiplied by the density of the medium. Thus the quantum behavior is accounted for entirely in the cross section, whereas the remaining propagation is treated classically. As shown previously (12, 13), absolute cross sections for elastic and inelastic scattering of an electron in amorphous ice can be obtained by performing a two-stream multiple-scattering analysis of the backscattered electron energy distribution, as provided by HREEL experiments. The main features of this two-stream model, developed previously (14) and applied to amorphous ice (12, 13), are summarized in this section.

Backscattered Electron Energy Distribution

For an incident electron beam of energy E_0 entering a disordered solid film, let us find the backscattered electron intensity, $J(E)$, which arises

from incoherent multiple elastic and inelastic events within the film, as a function of the final electron energy E . A simple phenomenological approach to this problem, when the detail of the angular distribution of $J(E)$ is not needed, is to solve the Boltzmann equation for plane-parallel systems in the “two-stream” approximation (14). In this approximation one simply considers the quantity $Q_r(E - E_0)$, defined as an angularly integrated scattering probability per unit length (SPUL) to lose or absorb an energy $E - E_0$. The possible angular dependence of a scattering event is described simply by splitting $Q_r(E - E_0)$ into two parts by introducing a coefficient of angular anisotropy $\gamma(E - E_0)$. Thus one defines an isotropic scattering part where the electron deflection occurs *either* in the forward or backward direction with the same probability, $Q_r(E - E_0) \equiv \frac{1}{2} [1 - \gamma(E - E_0)] Q_r(E - E_0)$, and an anisotropic part where the electron deflection arises *only* in the forward direction, $Q_r(E - E_0) \equiv \gamma(E - E_0) Q_r(E - E_0)$ (13). From a more physical point of view, the isotropic part is related to collisions arising from short-range interactions (e.g. resonant and ion-core scattering), whereas the anisotropic part is akin to long-range interactions (e.g. dipole scattering).

Considering the SPUL values constant within an energy interval equal to the energy-loss range investigated, the energy integrals in the transport equations reduce to a convolution form (14). This is practically the case for energy transfer to intermolecular and intramolecular vibrations, because the energy loss involved is usually much smaller than the primary beam energy. Owing to the convolution form, one may further introduce the Fourier transforms of the functions involved, such as

$$Q_{r,f}(E - E_0) = (1/2\pi) \int_{-\infty}^{\infty} ds Q_{r,f}(s) \exp[-is(E - E_0)], \quad (1)$$

and solve the transformed transport equations along with appropriate boundary conditions for the backscattered intensity as a function of the final energy E as

$$J^T(E) = (1/2\pi) \int_{-\infty}^{\infty} ds \frac{I_0(s) T R(s)}{1 - (1 - T) R(s)} \exp(-isE). \quad (2)$$

In this expression $I_0(s)$ is the Fourier transform of the incident current energy distribution entering the film. The transmission coefficient T accounts for the fraction of the outgoing electrons that are allowed by energy- and momentum-conservation laws (i.e. Snell-Descartes law) to overcome the potential barrier set up between the inner film potential V_p and the vacuum level at $E = 0$ (23). The vacuum level corresponds to an electron being at rest sufficiently far away from the film. Since the inner film potential V_p is usually attractive, it is considered negative with respect to the vacuum level (i.e. $V_p \leq 0$). In the case of an isotropic distribution of outgoing electrons having an energy $E > 0$, the transmission is given simply by

$$T = 1 - [|V_p|/(E + |V_p|)]^{1/2} \quad \text{for } E > 0. \quad (3a)$$

For outgoing electrons having an energy $V_p \leq E \leq 0$, which may occur for electrons propagating inside the film, the electrons are totally reflected by the potential barrier, and the transmission coefficient is

$$T = 0 \quad \text{for } V_p \leq E \leq 0. \quad (3b)$$

The term $R(s)$ is the Fourier transform of the electron energy distribution, $R(E - E_0)$, reflected from a film of thickness L . It is given explicitly (14) by

$$\begin{aligned} R(s) = & (R_\infty(M - 1/R_\infty) - (1/R_\infty)(M - R_\infty)) \\ & \times \exp\{-2\varepsilon\beta(s)L[(1 - R_\infty)/(1 + R_\infty)]\} \\ & \div \langle (M - 1/R_\infty) - (M - R_\infty) \rangle \\ & \times \exp\{-2\varepsilon\beta(s)L[(1 - R_\infty)/(1 + R_\infty)]\}, \end{aligned} \quad (4a)$$

with

$$R_\infty \equiv R_\infty(s) = \frac{1 - [1 - 2Q_r(s)/\beta(s)]^{1/2}}{1 + [1 - 2Q_r(s)/\beta(s)]^{1/2}} \quad (4b)$$

being the Fourier transform of the reflected energy distribution above, but in the limit of a semi-infinite film (i.e. $L \rightarrow \infty$). In both expressions, $\beta(s) \equiv \alpha - Q_r(s)$, where α is the total SPUL defined as

$$\alpha = \lambda^{-1} = \int_{-\infty}^{\infty} dE Q_r(E - E_0), \quad (5)$$

which also corresponds to the inverse of the total MFP λ , and $Q_r(s)$ is given by Eq. (1). Here ε is a phenomenological parameter that arises from considering an angular average in a stream direction. It has been found previously (12) to correlate simply to the average of the geometrical pathlength factor in the incident direction $1/\cos \theta_0$ and backscattered direction $1/\cos \theta_r$. The term M is the reflection coefficient of the underlying substrate for the condensed film and may include elastic as well as inelastic processes. However, its exact knowledge will turn out to be nonessential, as will be seen in the next section.

One can further see in Eq. (4b) that in the limit of large film thickness (i.e. $L \rightarrow \infty$) the overall reflectivity is no longer dependent on L , ε , as well as on the reflectivity of the substrate (i.e. M). More importantly and not obvious *a priori*, each SPUL is normalized to the total SPUL α in this expression. Consequently, as a result of multiple scattering processes within a semi-infinite film, a variation in a measured backscattered intensity may arise from a change of either the corresponding energy-loss SPUL value or the total SPUL α or both.

Method to Extract the Total Scattering Cross Section

In previous work (12), we developed and tested a procedure to extract the total SPUL α or total scattering cross section for an electron propagating within a film. The method relies essentially on the change in the elastic electron reflectivity as a function of the film thickness or more exactly the absolute molecular coverage.

The elastic electron reflectivity, by virtue of its zero-loss character, is a constant in the Fourier space s . The corresponding expression readily follows from Eq. (4), upon introduction of the elastic SPUL values according to $Q_r(s) \rightarrow Q_{re}$ and $Q_f(s) \rightarrow Q_{fe}$ along with $R_\infty(s) \rightarrow R_{\infty e}$, $\beta(s) \rightarrow \alpha - Q_{fe}$, and $M \rightarrow M_e$. Solving the resulting expression for the factor in the exponential yields

$$\ln \left[\frac{R_e/R_{\infty e} - 1}{M_e/R_{\infty e} - 1} \right] + \ln \left[\frac{1 - R_e R_{\infty e}}{1 - M_e R_{\infty e}} \right] = -2\varepsilon\alpha_{\text{eff}} \frac{L(1 - R_{\infty e})}{(1 + R_{\infty e})}, \quad (6)$$

where $\alpha_{\text{eff}} \equiv \alpha - Q_{fe}$ is defined as an effective total SPUL. The subtraction of Q_{fe} from α in the latter definition merely stems from the two-stream approach since small-angle *elastic* scattering cannot contribute to the electron-beam attenuation. Even if Q_{fe} and α cannot be found separately, the quantity α_{eff} can still be used along with the knowledge of Q_{re} to determine the total inelastic SPUL defined as $\alpha_i \equiv \alpha_{\text{eff}} - 2Q_{re}$. The factor $(1 - R_{\infty e})/(1 + R_{\infty e})$ on the right-hand side of Eq. (6), which in effect renormalizes α_{eff} accounts for multiple elastic scattering within the film. We note that for $R_{\infty e} < 0.1$, this factor turns out to be almost insensitive to a variation in $R_{\infty e}$. For instance, if $R_{\infty e}$ varies by 100% from 0.05 to 0.1, this factor changes by only 10% from 0.90 to 0.82. On the left-hand side of Eq. (6), the dependence of R_e on the film thickness L is contained in both terms. However, if we have $R_e R_{\infty e} \ll 1$ and $M_e R_{\infty e} \ll 1$, the second term on the left-hand side becomes close to zero and thus may safely be neglected in comparison to the first one. Hence the semi-logarithmic plot of the elastic reflectivity R_e normalized to that at infinite film thickness $R_{\infty e}$ minus one as function of the film thickness L should yield a straight line whose slope is directly proportional to α_{eff} . Finally, it should be emphasized further that under these conditions the determination of α_{eff} does not require the exact knowledge of the elastic reflectivity of the substrate M_e , because the latter appears only in the thickness-independent terms on the left-hand side of Eq. (6).

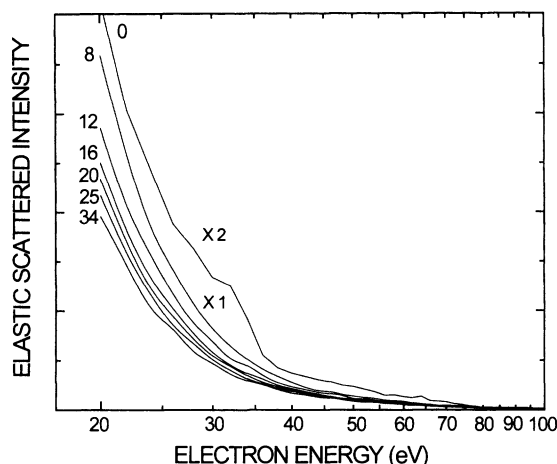


FIG. 1. Measurement of the elastic scattered electron intensity as function of the incident energy for several thicknesses of amorphous films of ice condensed at a temperature of 14 K on a polycrystalline platinum substrate. The angles of incidence, θ_0 , and analysis, θ_a , are 14° and 45° , respectively, with respect to the outward normal of the surface. The numbers near each curve indicate the thickness of the film in layers.

RESULTS

Elastic Electron Reflectivity

We previously reported the elastic electron reflectivity between 1.2 and 19.2 eV of water condensed at a temperature of 14 K on platinum for several film thicknesses (12). In general the reflectivity of the adlayers was found to exhibit an increase in the off-specular direction (i.e. $\theta_0 = 14^\circ$) up to about eight layers along with a large decrease in the specular direction (i.e. $\theta_0 = 45^\circ$). Above eight layers, the reflectivity was practically independent of the incident angle θ_0 . The decrease in the specular reflectivity was attributed to modifications in the film growth mode for the first few layers of water on the platinum substrate in going to the amorphous form of ice. Beyond eight layers the elastic reflectivity, which showed two broad maxima at 6 and 14.5 eV, was observed to decrease gradually with additional coverage.

The elastic reflectivity in the off-specular direction of a similarly grown water film is shown from 20 to 100 eV in Fig. 1. The curve labeled 0 is the corresponding elastic reflectivity of the platinum substrate shown with a magnification of $2\times$. As in our previous measurement, the reflectivity reaches a maximum at about eight layers and decreases for higher coverage. The lower reflectivity below eight layers is observed along with a higher reflectivity in the specular direction. In a sense, the higher specular reflectivity can be seen as arising at the expense of the off-specular intensity. Beyond eight layers, the reflectivity exhibits a monotonous decrease from 20 to 100 eV. As before, the sensitivity of the present instrument to film charging (e.g. electron trapping and hole creation) limited measurements for thicknesses less than 40 layers. It should be pointed out, since the present method to extract the total scattering cross section relies on relative reflectivity values, that

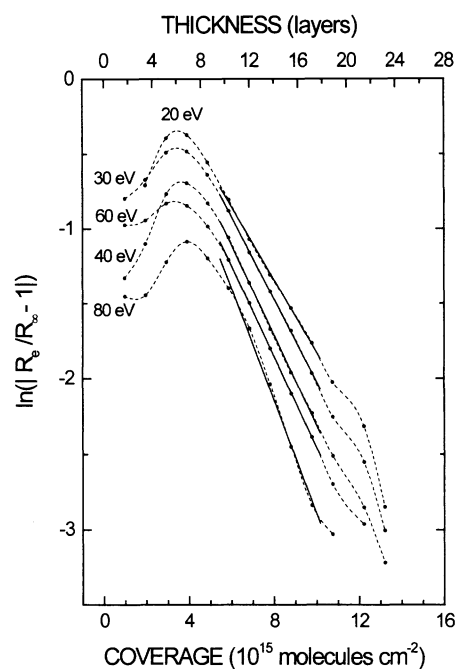


FIG. 2. Semi-logarithmic plot of the normalized elastic electron reflectivity minus one as a function of film thickness for various incident electron energies. The normalization constant for each energy is taken as corresponding to the elastic reflectivity at 34 layers. The latter proves to be reasonably close to the elastic reflectivity of the semi-infinite film R_{∞}^e . Each straight line results from a least-squares-fit analysis of the data comprised between 10- and 18-layer coverage (continuous part), and its slope is related to the effective total and total inelastic electron scattering cross section per scatterer.

an exact knowledge of the electron transmission of the spectrometer optics is not required.

Total Scattering Cross Section

For the few incident energies selected in Fig. 1, we present in Fig. 2 the semi-logarithmic plots of the elastic reflectivity ratio as prescribed by Eq. (6) with the 34-layer curve being used as the semi-infinite data, R_{∞}^e . For all energies, a linear behavior is observed from 10 to about 18 layers. The departure from the straight line at low coverage is partly due to the presence of an intense specular component that merely depletes some intensity from the off-specular component, as mentioned above. Deviation from the linear behavior and scatters in the data are also expected as the film thickness approaches that chosen for the reflectivity normalization R_{∞}^e . The slope (b) of a straight line is related to the total SPUL α_{eff} with

$$b = -2\epsilon\alpha_{eff}(1 - R_{\infty}^e)/(1 + R_{\infty}^e). \quad (7)$$

Since the film thickness is actually determined in terms of absolute molecular coverage corresponding to the product $n \times L$, where n is the bulk density of the sample, the slope is more exactly related to the effective total scattering cross section per scatterer $\sigma_{T,eff}$, which is defined relatively to the effective total SPUL α_{eff} by $\sigma_{T,eff} \equiv \alpha_{eff}/n$.

The energy dependence of the effective total scattering

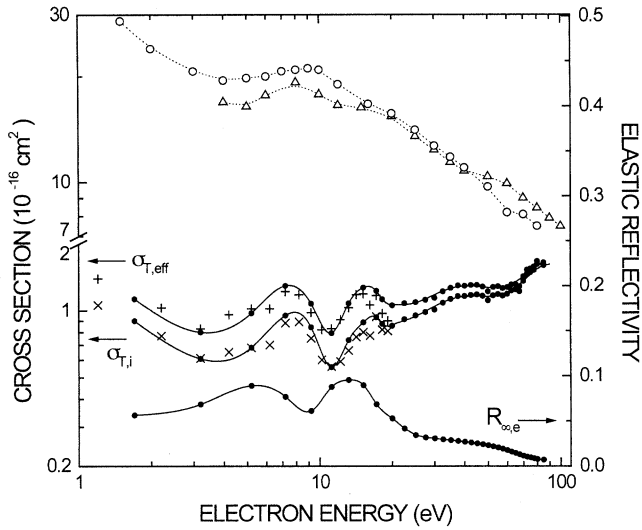


FIG. 3. The closed-symbol continuous curve at the top is the effective total cross section per scatterer $\sigma_{T,eff}$ for electron scattering in amorphous ice. The closed-symbol continuous curve at the bottom is the intrinsic (i.e. without the effect of the film-vacuum potential barrier) elastic electron reflectivity R_{∞} of a semi-infinite film of amorphous ice (see text). It is related to the isotropic elastic scattering cross section by Eq. (8). The knowledge of the latter cross section along with the value of $\sigma_{T,eff}$ through Eq. (9) yields the total inelastic-scattering cross section per scatterer $\sigma_{T,i}$ shown as the closed-symbol continuous curve in the middle. The crosses correspond to our previous cross-section data (12, 13); +, for $\sigma_{T,eff}$; $\times$, for $\sigma_{T,i}$. The open symbols correspond to the total cross section for electron scattering on water in the gas phase; $\circ$, Szmytkowski (25); $\triangle$, Saglam and Aktekin (26, 27).

cross section $\sigma_{T,eff}$ is shown between 1 and 100 eV in Fig. 3. Each point corresponds to the slope b extracted at the given energy (e.g. Fig. 2) after corrections for the ϵ parameter and the energy-dependent elastic multiple-scattering factor, $(1 - R_{\infty})/(1 + R_{\infty})$, as prescribed by Eq. (7). The ϵ parameter, which depends on the experimental scattering geometry, was chosen to be constant in energy and equal to 1.2 as in our previous work (12, 13). The elastic multiple-scattering factor was calculated with the intrinsic elastic reflectivity R_e , which is shown in the bottom of Fig. 3. This reflectivity is obtained from the Eq. (4b), which is evaluated at the elastic SPUL points $Q_r(s) \rightarrow Q_{re}$ and $Q_f(s) \rightarrow Q_{fe}$, i.e.

$$R_{\infty} = \frac{1 - [1 - 2Q_{re}/\alpha_{eff}]^{1/2}}{1 + [1 - 2Q_{re}/\alpha_{eff}]^{1/2}}, \quad (8)$$

given the experimental value of the normalized elastic SPUL $2Q_{re}/\alpha_{eff}$. The latter ratio is determined independently from the two-stream analysis of the corresponding energy-loss spectrum, as described in detail below. The curve just below $\sigma_{T,eff}$ in Fig. 3 is the total inelastic-scattering cross section $\sigma_{T,i} \equiv \alpha_i/n$. It is readily obtained from the definitions $\alpha_i \equiv \alpha_{eff} - 2Q_{re}$ and $\sigma_{T,eff} \equiv \alpha_{eff}/n$ through the following relationship:

$$\sigma_{T,i} = \sigma_{T,eff}(1 - 2Q_{re}/\alpha_{eff}), \quad (9)$$

given again the value of the normalized elastic SPUL $2Q_{re}/$

α_{eff} . The present $\sigma_{T,eff}$ and $\sigma_{T,i}$ set differs from our previous one in the region of overlap, displayed as crosses in Fig. 3, by showing only a small relative increase in the cross-section values with the electron energy. The same general trend is observed for the elastic and all inelastic scattering cross sections as they are obtained by multiplying their associated normalized SPUL by the same effective total scattering cross section.

In Fig. 3 the scatter in the data reflects only a smoothed root-mean-square deviation among the slope values. The relative uncertainty in the raw data amounts to $\pm 5\%$ between 1 and 20 eV and must be considered to increase progressively up to $\pm 20\%$ at 100 eV. It does not include the systematic uncertainty arising from the calibration of the molecular coverage (i.e. nL) at $\pm 15\%$, the ϵ parameter, or the elastic multiple-scattering factor. Considering a combined error of $\pm 10\%$ for the latter two quantities, as determined previously from the fit to the data generated by a Monte Carlo simulation (12), a total systematic uncertainty of $\pm 25\%$ is obtained for the cross-section scale. Adding to the latter uncertainty, the above relative uncertainty in the raw data yields an overall error of ± 30 – 45% .

The numerical values of the total scattering cross section per scatterer $\sigma_{T,eff}$, $\sigma_{T,i}$ and the corresponding computed MFP values, obtained upon using the bulk density of low-density amorphous ice at 3.15×10^{22} molecule cm^{-3} (20), are reported in Table 1. Also reported are the electron attenuation length (AL) values of Kurtz *et al.* (24) measured between 18 and 68 eV on a similarly grown ice film at a temperature of 90 K. Their AL values compare, within experimental uncertainties, to our corresponding MFP values in Table 1, $|2/b| = 1/\epsilon n \sigma_{T,eff} (1 - R_{\infty})/(1 + R_{\infty})$, which do not include correction for the ϵ parameter and the energy-dependent elastic multiple-scattering factor.

The open symbols in Fig. 3 correspond to the total cross sections for electron scattering on water that have been measured in the gas phase (25–27) and that are covering the present energy range. These two representative sets will be discussed later.

Vibrational Energy-Loss Spectra

The water molecule in its ground state is bent and belongs to the C_{2v} point group (28). The C_2 axis of rotation defines the z direction; the plane of the molecule lies in the xz plane, and the y direction is perpendicular to the plane of the molecule. The vibrations of water in the gas phase are the bending mode (ν_2) found at 198 meV and the OH stretching modes divided into a symmetrical stretch (ν_1) at 453 meV and an asymmetrical stretch (ν_3) at 466 meV (29). Upon condensation into amorphous (i.e. vitreous) ice, all water molecules are equivalent in a tetrahedral geometry with a mean distance of 2.76 Å between the oxygen atoms (20). The water molecule acts as a donor as well as an acceptor with the oxygen atoms covalently bonded to two

TABLE 1
Total Effective $\sigma_{T,eff}$ and Total Inelastic $\sigma_{T,i}$ Cross
Sections per Scattered (10^{-16} cm²) Along with the
Corresponding $1/n\sigma_{T,eff}$ and $1/ns_{T,i}$ MFP Values ($\text{\AA}$)
for Electron Scattering in the Low-Density
Amorphous Form of Ice at 14 K

E_0 (eV)	$\sigma_{T,eff}$	$1/n\sigma_{T,eff}$	$\sigma_{T,i}$	$1/ns_{T,i}$	$ 2/b $	AL ^a
1.7	1.13	28.1	0.90	35.2	26.2	
3.2	0.80	39.7	0.61	52.2	38.0	
5.2	0.97	32.6	0.68	46.6	32.5	
7.2	1.30	24.5	0.95	33.3	23.8	
9.2	1.08	29.5	0.84	37.7	27.8	
11.2	0.79	40.1	0.56	57.0	39.8	
13.2	1.08	29.4	0.74	43.1	29.7	
15.2	1.27	25.0	0.89	35.8	24.9	
17.2	1.23	25.9	0.93	34.0	24.7	
18.2	1.13	28.2	0.87	36.4	26.7	18.2 (18 eV)
20	1.06	30.1	0.85	37.1	27.8	
22.5	1.08	29.4	0.91	34.7	26.6	
25	1.09	29.1	0.95	33.4	26.0	
27.5	1.12	28.4	0.98	32.2	25.2	13.6 (28 eV)
30	1.14	27.8	1.01	31.4	24.6	
32.5	1.22	26.0	1.08	29.3	23.0	
35	1.27	25.1	1.13	28.2	22.1	
37.5	1.30	24.4	1.16	27.3	21.5	16.4 (38 eV)
40	1.29	24.5	1.16	27.3	21.6	
42.5	1.30	24.3	1.17	27.0	21.4	
45	1.31	24.3	1.18	26.9	21.3	
47.5	1.28	24.7	1.17	27.2	21.6	16.8 (48 eV)
50	1.22	26.0	1.11	28.5	22.7	
52.5	1.28	24.9	1.17	27.1	21.6	
55	1.29	24.6	1.19	26.8	21.4	
57.5	1.26	25.2	1.17	27.2	21.8	16.4 (58 eV)
60	1.28	24.9	1.19	26.7	21.5	
62.5	1.31	24.2	1.23	25.8	20.8	
65	1.32	24.1	1.24	25.5	20.6	
67.5	1.29	24.7	1.22	26.0	21.1	16.8 (68 eV)
70	1.43	22.3	1.36	23.3	19.0	
72.5	1.53	20.8	1.47	21.6	17.7	
75	1.55	20.5	1.49	21.3	17.4	
77.5	1.58	20.1	1.52	20.8	17.1	
80	1.67	19.0	1.62	19.6	16.1	
85	1.65	19.2	1.60	19.8	16.3	
90	1.52	20.9	1.48	21.5	17.7	
100	1.50	21.2	1.46	21.8	17.9	

Notes. MFP values ($\text{\AA}$), $|2/b| = 1/\epsilon\sigma_{T,eff} (1 - R_{\infty})/(1 + R_{\infty})$, which are obtained without correction for the ϵ parameter and multiple elastic scattering. Electron attenuation lengths AL values ($\text{\AA}$) in solid H₂O condensed at 90 K.

^a Electron attenuation lengths values were taken from ref. (24) at the energy (eV) in parentheses.

hydrogen atoms and hydrogen bonded to two other hydrogen atoms.

The vibrational energy-loss spectra of a 30-layer film of water condensed at a temperature of 14 K is shown for several incident electron energies between 20 and 100 eV in Fig. 4. The spectra below 20 eV were reported for the same film thickness and scattering angle in our previous work (13). The dotted curves correspond to the experimental recordings, while the solid curves result from the multiple-scattering calculation to be discussed in the next sec-

tion. As in our previous work, the 30-layer film is a good balance between having the backscattered intensity close to that of the semi-infinite solid while keeping film charging small.

From the comparison with the infrared and Raman data, the most prominent energy-loss peaks are attributed to the excitation of intermolecular and intramolecular fundamental vibrational modes of amorphous ice. The large peak near zero energy loss ($E_0 - E = 0$ eV) arises from the electrons scattered elastically and quasielastically. Its asymmetrical shape along with the shoulder at 25–26 meV is the result of excitation of the lattice phonon (ν_T) modes. In the far-infrared spectra of cubic and hexagonal ice (30), these modes are observed as two bands at about 20 and 29 meV. The relatively large maximum near 90 meV, which was previously found to separate into two bands centered around 65 and 95 meV at 9.2 eV incident energy (13), decreases gradually in intensity at higher incident energy and becomes merely a shoulder above 50 eV. These bands are attributed to different librational phonon modes (ν_L) that derived from the three frustrated rotations of the water molecules. Their large observed width is due to the crystal field and hydrogen bonding (31). Similarly to the infrared and Raman spectra of ice (31, 32), we ascribe the 65 meV feature to the superposition of molecular librations around the y and z directions and the 95 meV feature to that in the x direction. The narrow peak at 205 meV is assigned to the bending mode ν_2 , which becomes stiffer because of the fairly directional hydrogen bonding (33). The broad maximum at about 417 meV, which appears within the 420–430 meV range below 15 eV incident energy, is attributed to the stretching modes $\nu_{1,3}$ of the water molecule. Again owing to hydrogen bonding, these vibrations are shifted to lower energy compared to the gas phase and broadened as a result of the spreading out of the bond restoring force constant with local disorder (34). Finally, the small shoulder at 460 meV, whose energy corresponds closely to the OH stretch ν_3 in the gas phase, is attributed to the presence of free OH oscillators not involved in hydrogen bonds at a discontinuity, as for instance at the film surface (35).

Integral Elastic and Inelastic Cross Sections

Because of the spread of the vibrational restoring force constant in amorphous ice, the various vibrational SPUL values are taken as a sum of Gaussian energy distribution functions centered around a mean vibrational excitation energy $h\nu_i$. Since $\alpha_{eff} \equiv \alpha - Q_{fe}$ is actually the quantity determined experimentally, the various SPULs that are normalized to α in Eq. (4) may be expressed for convenience as

$$Q_i(E - E_0)/\alpha_{eff} = \sum_i a_i \exp[-(E - E_0 + h\nu_i)^2/b_i^2] \quad (10a)$$

with

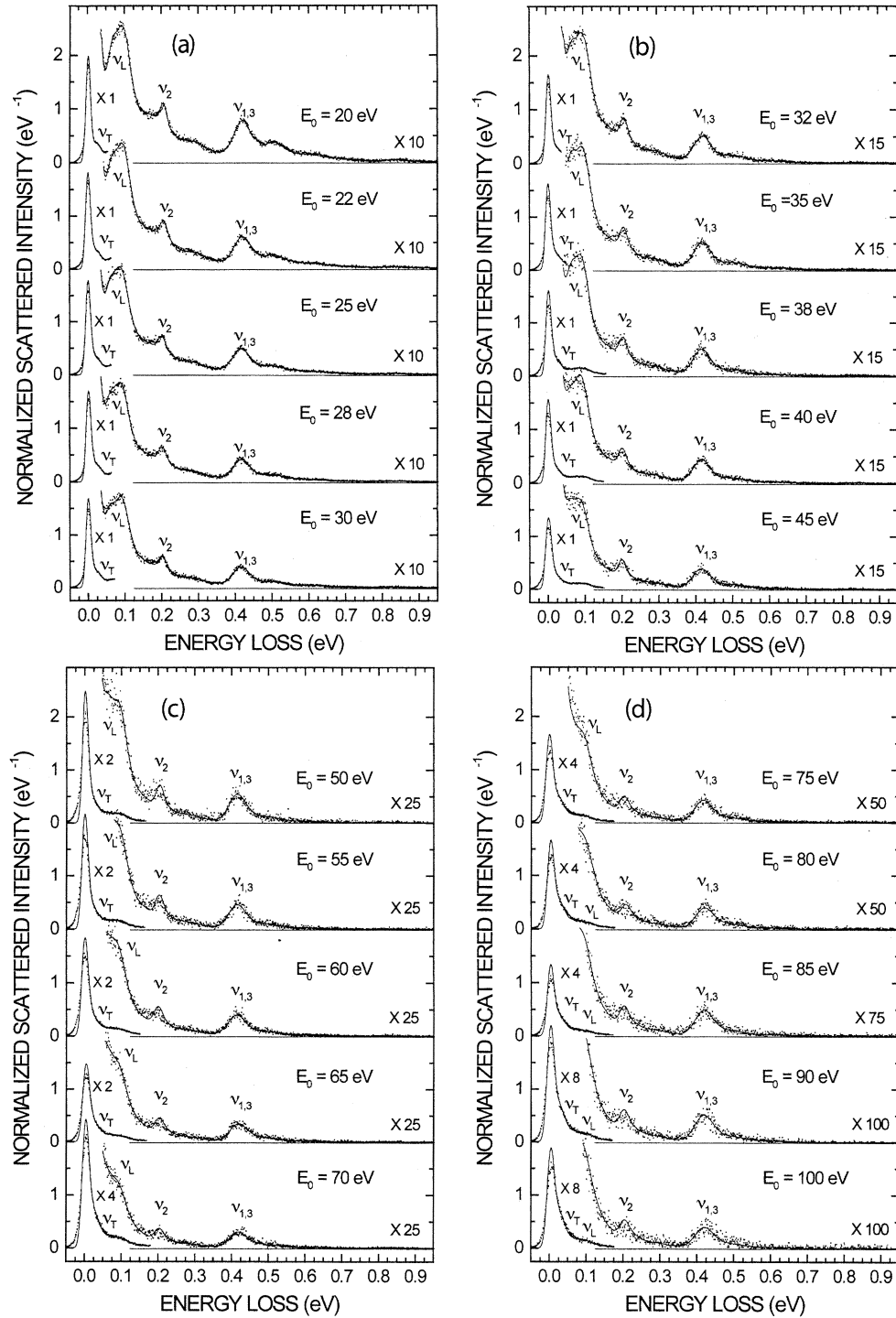


FIG. 4. Energy-loss spectra of 30-layer films of ice condensed at a temperature of 14 K on a polycrystalline platinum substrate for several incident electron energies between (panel a) 20 and 30 eV, (panel b) 32 and 45 eV, (panel c) 50 and 70 eV, and (panel d) 75 and 100 eV. For each energy, the dotted curve is the normalized experimental energy-loss spectrum, which was recorded at an angle of incidence $\theta_0 = 14^\circ$ and analysis $\theta_d = 45^\circ$. The superimposed continuous curve is the calculated backscattered electron energy distribution resulting from the two-stream analysis. With the present intensity scale, the integral of an elastic or inelastic feature can be expressed in terms of an absolute reflectivity.

$$b_i = \delta_i / [2(\ln 2)^{1/2}], \quad (10b)$$

where a_i is the amplitude and δ_i the FWHM of a distribution i . The sum over a given distribution is given analytically by

$$Q_{ti} / \alpha_{eff} = a_i b_i \pi^{1/2}, \quad (11)$$

and it represents the angularly integrated SPUL pertaining to the vibrational mode i of mean excitation energy $h\nu_i$. The angular dependence of a scattering event is accounted

for simply through a coefficient of angular anisotropy ($0 \leq \gamma_i \leq 1$) by splitting the SPUL distribution (Eq. 10a) into an isotropic and anisotropic components, respectively:

$$\frac{Q_r(E - E_0)}{\alpha_{eff}} = \sum_i \frac{1}{2} (1 - \gamma_i) a_i \exp \left[-\frac{(E - E_0 + hv_i)^2}{b_i^2} \right] \quad (12a)$$

$$\frac{Q_f(E - E_0)}{\alpha_{eff}} = \sum_i \gamma_i a_i \exp \left[-\frac{(E - E_0 + hv_i)^2}{b_i^2} \right] \quad (12b)$$

The value of the relative elastic SPUL (i.e. Q_r/α_{eff}) is first estimated from the intensity of the experimental elastic peak upon using Eq. (8). Then follows the calculation of the backscattered energy distribution as prescribed by Eqs. (2)–(4) with $I_0(s)$ taken as a unit-normalized Gaussian distribution of the FWHM equal to the finite energy resolution of the spectrometer (i.e. 15–20 meV FWHM). Normally, the calculations begin by first adjusting the elastic peak and then proceeding toward higher energy loss with the introduction of just one SPUL distribution at a time. The mean energy and the width of each SPUL distribution are initially inferred from the experimental spectrum. During the course of the analysis, these parameters are refined further as a result of a close comparison between the experimental and calculated data. What one usually finds is that some experimental spectral features can be accounted for only with additional SPUL values, whereas others stem from multiple scattering effects. In the present calculations, the backscattered energy distribution is generated with an energy mesh of 1 meV, the same as the experimental spectrum. Also V_p , which accounts for the long-range part of the interaction potential inside the film smoothly joining the image potential outside, is fixed to the value of the bottom of the lowest conduction band V_0 at 1 eV below the vacuum level. The latter value is actually deduced from the difference between the electron photoconductivity and photoemission thresholds in ice at 2.3 eV (36) and 3.3 eV (37), respectively.

In Fig. 4, we present the calculated energy distribution curves (solid line) superimposed on their respective experimental spectra (dotted line). The fit between each calculated and experimental integrated intensities as a function of the energy loss was accurate within 5%. The apparent disagreement between the measured and calculated elastic intensity results from the shape of the experimental elastic distribution being larger than a Gaussian function at the base. In this case, only the integral of the elastic peak was considered for the comparison with no attempt to fit the shape exactly. Otherwise all inelastic features in a spectrum could be reproduced with the set of parameters as defined above.

The parameters for each of the calculated curves are compiled for all energies in Tables 2 and 3. The relative SPUL values obtained for the elastic scattering and the various excitation modes are included scaled to the effective total cross section per scatterer (i.e. $\sigma_{T,eff} \equiv \alpha_{eff}/n$), which

was determined in Table 1. In particular, the energy dependence of the various integral cross sections is shown in Figs. 5 and 6. In principle, the cross sections thus obtained are free from superimposition among the various energy-loss distributions as well as the presence of incoherent multiple scattering, which appears as the molecular coverage increases. However, besides the quantum interferences pertaining to the scattering on a single water molecule, they may still include additional interference effects along with screening phenomena due to the presence of neighboring molecules. These latter effects are discussed in the next section.

DISCUSSION

Electron Scattering in Condensed Matter

Low-energy electron scattering in the gas phase is described microscopically by the sum of the scattered *intensities* originating from individual atoms or molecules. The same is not true in the condensed phase for two different reasons. The first stems from the quantum-mechanical nature of the electron, whose effect becomes conspicuous when the energy of the electron or more exactly its de Broglie wavelength compares, within an order of magnitude, to the closest distance separating the atomic or molecular sites. As a result, one must include interference as well as coherent multiple-scattering contributions among the scattering sites in addition to the individual site contribution. The second reason comes into play when the interactions of an atomic or a molecular site with its neighbors can no longer be neglected. These perturbations normally lead to the modification of the ground state of an atom or molecule, owing to the electrostatic and van der Waals interactions, along with the formation of various excitons in a medium (i.e. nuclear and electronic excitation modes) (38). Also, since the latter excitation modes are generally delocalized, they allow for quantum interference effects in an inelastic scattering process (i.e. conservation of momentum transfer). More importantly, the perturbation among the sites is responsible for the screening of the interaction of an electron with an atom or molecule by its neighbors. This additional screening has an immediate effect on the strength of the elastic and inelastic scattering that would originate from noninteracting sites (i.e. isolated targets). In the following, the energy dependence and magnitude of the present cross sections is discussed in terms of its isolated-site contribution, which is similar to the gas phase, and multiple-site contribution along with interferences and screening effects due to the presence of neighboring sites.

1. Isolated-site contribution

It is most convenient to discuss electron scattering from an isolated site in terms of direct (39) and resonant processes (40, 41) as in the gas phase. Direct processes responsible for the excitation of the vibrational, rotational and

TABLE 2
Integral Cross Section per Scatterer as a Function of the Incident Energy E_0 (eV) for Electrons Elastically Scattered and Exciting Various Intermolecular Vibrational Modes in Amorphous Ice at 14 K

E_0 (eV)	Excitation modes									
	Elastic		ν'_T		ν''_T		ν'_L		ν''_L	
	σ	γ	σ	γ	σ	γ	σ	γ	σ	γ
1.7	0.229	0	0		0.203	0.3	0.120	0.55	0.485	0.79
3.2	0.192	0	0		0.146	0.39	0.080	0.68	0.295	0.81
5.2	0.293	0	0		0.157	0.25	0.085	0.66	0.311	0.83
7.2	0.343	0	0		0.199	0.35	0.125	0.66	0.412	0.91
9.2	0.234	0	0.006	0	0.120	0.35	0.126	0.66	0.315	0.94
11.2	0.235	0	0.001	0	0.083	0.35	0.091	0.66	0.204	0.94
13.2	0.343	0	0.017	0	0.117	0.4	0.106	0.7	0.242	0.96
15.2	0.384	0	0.018	0	0.137	0.5	0.121	0.8	0.265	0.98
17.2	0.295	0	0.015	0	0.141	0.63	0.117	0.87	0.246	0.98
20	0.201	0	0.012	0	0.120	0.68	0.105	0.91	0.207	0.98
22.5	0.166	0	0.013	0	0.125	0.72	0.108	0.92	0.199	0.99
25	0.141	0	0.013	0	0.124	0.75	0.111	0.96	0.191	0.99
27.5	0.134	0	0.016	0	0.125	0.75	0.115	0.96	0.180	1
30	0.132	0	0.016	0	0.127	0.75	0.118	0.98	0.171	1
32.5	0.137	0	0.019	0	0.131	0.75	0.124	0.98	0.175	1
35	0.138	0	0.021	0	0.130	0.75	0.126	0.98	0.175	1
37.5	0.138	0	0.024	0	0.130	0.75	0.127	0.98	0.170	1
40	0.133	0	0.026	0	0.127	0.76	0.123	1	0.160	1
42.5	0.130	0	0.027	0	0.124	0.77	0.121	1	0.154	1
45	0.125	0	0.026	0	0.119	0.77	0.120	1	0.145	1
47.5	0.119	0	0.025	0	0.115	0.78	0.117	1	0.135	1
50	0.108	0	0.023	0	0.108	0.79	0.109	1	0.124	1
52.5	0.107	0	0.024	0	0.111	0.79	0.112	1	0.126	1
55	0.102	0	0.024	0	0.110	0.79	0.112	1	0.124	1
57.5	0.094	0	0.023	0	0.108	0.79	0.110	1	0.120	1
60	0.087	0	0.024	0	0.109	0.79	0.112	1	0.119	1
62.5	0.081	0	0.024	0	0.112	0.79	0.118	1	0.121	1
65	0.074	0	0.023	0	0.113	0.79	0.121	1	0.120	1
67.5	0.065	0	0.020	0	0.110	0.79	0.120	1	0.116	1
70	0.064	0	0.019	0	0.121	0.79	0.132	1	0.128	1
72.5	0.062	0	0.016	0	0.130	0.79	0.142	1	0.138	1
75	0.058	0	0.012	0	0.130	0.83	0.143	1	0.139	1
77.5	0.054	0	0.009	0	0.130	0.83	0.144	1	0.141	1
80	0.051	0	0.007	0	0.136	0.85	0.152	1	0.148	1
85	0.046	0	0.006	0	0.132	0.86	0.150	1	0.145	1
90	0.040	0	0.005	0	0.119	0.87	0.136	1	0.132	1
100	0.037	0	0.004	0	0.117	0.87	0.134	1	0.129	1

Notes. ν'_T , translational mode at about 10 meV and FWHM of 1 meV; ν''_T , translational mode at about 24 meV and FWHM of 25 meV; ν'_L , librational mode at about 61 meV and FWHM of 30 meV; ν''_L , librational mode at about 92 meV and FWHM of 40 meV; σ , cross section per scatterer (10^{-16} cm²); γ , coefficient of angular anisotropy.

translational modes of a molecule result simply from the dependence of the effective electron–molecule interaction potential on the corresponding coordinate. This potential is described asymptotically by a single-center expansion into dipole, quadrupole and polarization terms, etc. Typically the cross section that one calculates for each of these terms exhibits a smooth behavior of as a function of the electron energy (42).

Direct scattering processes for the vibrational excitation of water have been calculated from threshold up to 100 eV within the first Born approximation by Itikawa (43). In that work, the dipole and polarization terms are found to contribute mainly to the cross sections below 10 eV, gradually reaching a maximum around 1 eV. The quadrupole contri-

bution is more nearly flat and dominates above 10 eV. Owing to the long-range character of the dipole and polarization interactions, the resulting differential cross sections are more pronounced toward small scattering angles. Also the angular spreading of the differential cross sections become narrower with increasing incident energy. On the other hand, the quadrupole term is contributing to a more uniform distribution. Still, the rotational excitations result from a torque generated by the dependence of the effective electron–molecule interaction potential on the target orientation. For the water molecule, which has a strong dipole moment, calculations of the electron scattering cross section within the first Born approximation show that the $00 \rightarrow 10$ rotational transition is by far the dominant excitation

TABLE 3
Integral Cross Section per Scatterer as a Function of the Incident Energy E_0 (eV) for Electrons Exciting
Various Intermolecular Vibrational Modes in Amorphous Ice at 14 K

E_0 (eV)	Excitation modes										
	ν_2		$\nu_{1,3}$		ν_3		$\nu_{1,3} + \nu_L$		$2(\nu_{1,3})$		Others
	σ	γ	σ	γ	σ	γ	σ	γ	σ	γ	σ
1.7	0.017	0	0.068	0	0.002	0	0.005	0	0	0	0.001
3.2	0.017	0	0.048	0	0.004	0	0.011	0	0.006	0	0
5.2	0.026	0.45	0.058	0	0.005	0	0.015	0	0.005	0	0.020
7.2	0.043	0.6	0.130	0.44	0.005	0	0.022	0.2	0.014	0	0.002
9.2	0.037	0.7	0.120	0.5	0.005	0	0.017	0.3	0.021	0	0.078
11.2	0.025	0.78	0.069	0.5	0.003	0	0.004	0.3	0.009	0	0.068
13.2	0.035	0.9	0.065	0.5	0.002	0	0.004	0.3	0.006	0	0.141
15.2	0.039	0.95	0.065	0.55	0		0.002	0.3	0.004	0	0.235
17.2	0.037	0.96	0.054	0.6	0		0.001	0.35	0.003	0	0.320
20	0.027	0.98	0.036	0.6	0		0.001	0.35	0.002	0	0.345
22.5	0.026	0.98	0.029	0.6	0		0		0.001	0	0.413
25	0.024	0.99	0.022	0.6	0		0		0		0.464
27.5	0.024	0.99	0.018	0.52	0		0		0		0.506
30	0.024	0.99	0.016	0.5	0		0		0		0.537
32.5	0.025	0.99	0.016	0.63	0		0		0		0.595
35	0.025	0.99	0.015	0.55	0		0		0		0.634
37.5	0.026	1	0.015	0.45	0		0		0		0.670
40	0.026	1	0.015	0.52	0		0		0		0.686
42.5	0.026	1	0.014	0.52	0		0		0		0.709
45	0.025	1	0.015	0.67	0		0		0		0.730
47.5	0.025	1	0.016	0.67	0		0		0		0.732
50	0.023	1	0.016	0.75	0		0		0		0.710
52.5	0.024	1	0.017	0.78	0		0		0		0.757
55	0.024	1	0.017	0.8	0		0		0		0.775
57.5	0.023	1	0.017	0.81	0		0		0		0.767
60	0.024	1	0.017	0.83	0		0		0		0.784
62.5	0.024	1	0.018	0.84	0		0		0		0.814
65	0.024	1	0.018	0.86	0		0		0		0.826
67.5	0.024	1	0.017	0.86	0		0		0		0.814
70	0.026	1	0.019	0.87	0		0		0		0.915
72.5	0.027	1	0.020	0.88	0		0		0		0.993
75	0.028	1	0.021	0.9	0		0		0		1.019
77.5	0.028	1	0.021	0.91	0		0		0		1.051
80	0.029	1	0.021	0.92	0		0		0		1.129
85	0.028	1	0.021	0.92	0		0		0		1.122
90	0.027	1	0.019	0.93	0		0		0		1.040
100	0.026	1	0.018	0.95	0		0		0		1.031

Notes. ν_2 , bending mode at about 204 meV and FWHM of 16 meV; $\nu_{1,3}$, stretching modes at about 417 meV and FWHM of 50 meV; ν_3 , asymmetrical stretch mode at about 460 meV and FWHM of 5 meV; $\nu_{1,3} + \nu_L$, stretching-librational combination mode at about 500 meV and FWHM of 40 meV; $2(\nu_{1,3})$ overtone of the stretching modes at about 835 meV and FWHM of 75 meV. Others correspond to residual vibrational excitations above 1 eV energy loss, dissociative attachment, electronic excitation as well as ionization processes; σ , cross section per scatterer (10^{-16} cm²); γ , coefficient of angular anisotropy.

(44, 45). The integrated cross section is found to decrease monotonously as a function of the incident energy up to 50 eV and presumably beyond. The corresponding differential cross sections as well as those extracted from experiments (46) are strongly peaked toward small scattering angles. It should be noted that within the framework of the adiabatic approximation, where the molecular rotation time is much larger than the electron collision time, the integrated cross section summed over all final rotational states is independent of the initial rotational state (45). Hence, if the elastic peak and the rotational transitions cannot be resolved, the measured cross section, which is given by the average over

all thermally accessible initial states, is found to be independent of the gas temperature. However, since each state-to-state cross section depends on the specific initial and final rotational states, the cross section for elastic scattering alone is expected to vary with the temperature along with that for rotational excitations varying in the opposite direction.

Electron resonance scattering or formation of a transient anion state occurs whenever an incident electron is captured into an empty orbital of an atom or a molecule during a time longer than the normal scattering time (40, 41, 47, 48). In electron scattering from the water molecule, the cap-

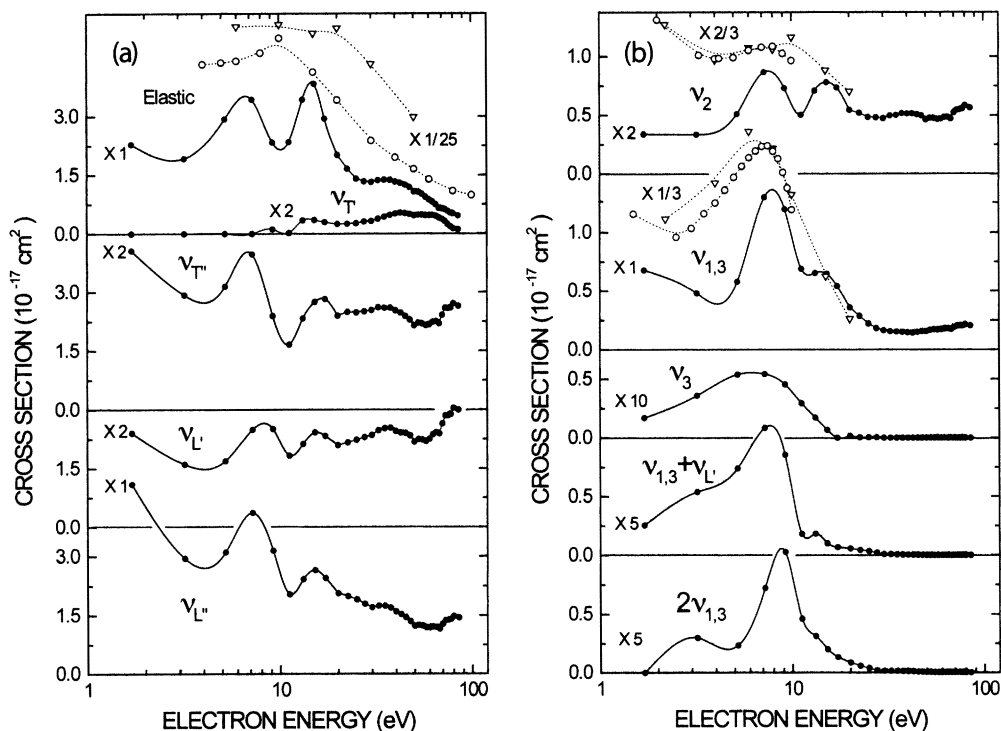


FIG. 5. Integral scattering cross sections per scatterer for (panel a) the elastic electron scattering and of the ν_T translational mode at about 10 meV, ν_T translational mode at about 24 meV; ν_L librational mode at about 61 meV and ν_L' librational mode at about 92 meV and (panel b) electron excitation of the ν_2 bending mode, $\nu_{1,3}$ stretching modes, ν_3 asymmetrical stretch mode; $\nu_{1,3} + \nu_L$ stretching-librational combination mode, and $2(\nu_{1,3})$ overtone of the stretching modes in amorphous ice as derived from the two-stream analysis of the energy-loss spectra of Fig. 4. Measured integral cross section for electron scattering from water in the gas phase (panel a) elastic scattering; $\cdots\circ\cdots$, Danjo and Nishimura (55); $\cdots\nabla\cdots$, Johnstone and Newell (53), and (panel b) excitation of the bending and stretching vibrational modes; $\cdots\circ\cdots$, Seng and Linder (50); $\cdots\nabla\cdots$, Shyn, Sho and Craven (52).

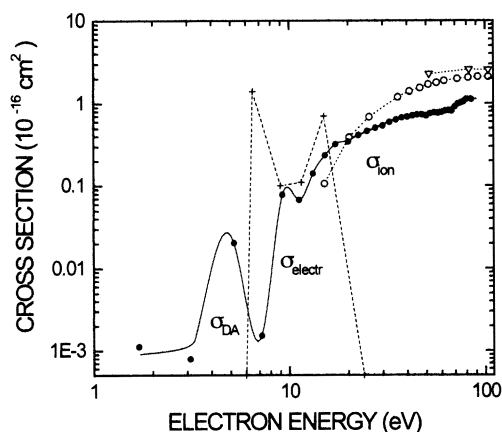


FIG. 6. Integral cross section ascribed to the sum of dissociative attachment (DA), vibrational excitation above 1 eV energy loss, electronic excitations (electr), as well as ionization (ion) processes in amorphous ice. The numerical value of this integral cross section is listed as “Others” in Table 3. Electron scattering cross section reported for electronic excitations of water in the gas phase; $\cdots + \cdots$, from Ness and Robson (86). Measured total electron impact ionization cross section for water in the gas phase; $\cdots\circ\cdots$, Djuric *et al.* (89) $\cdots\nabla\cdots$, Bolorizadeh and Rudd (90).

ture of an incident electron into the lowest vacant anti-bonding orbital $4a_1$ or $2b_2$ leads to the formation of a 2A_1 or 2B_2 anion state. The molecular electronic structure calculation of Claydon *et al.* (49) suggests that the A_1 state should be within an energy range of 2–3 eV and the 2B_2 state within an energy range of 6–7 eV. Electron impact elastic and vibrational (i.e. bending and stretching modes) cross-section measurements in the gas phase typically exhibit a strong enhancement at threshold (50, 51), a broad maximum in the 6–8 eV energy range (50, 52), and a monotonous decrease at higher energies (53–56). By comparing the cross-section calculations for the different symmetry species of the molecule to the measurements of Seng and Linder (50), Jain and Thompson (57, 58) were able to identify the contribution due to the B_2 state around 7 eV and that due to the A_1 state below 2 eV.

The lowest core-excited resonances of water are formed when two electrons occupy the first $3s4a_1$ orbital around the positive cores $\dots(1b_2)^2(3a_1)^2(1b_1)\dots$, $\dots(1b_2)^2(3a_1)(1b_1)^2$, or $\dots(1b_2)(3a_1)^2(1b_1)^2$, resulting in the 2B_1 , 2A_1 or 2B_2 anion states, respectively. In the gas phase, these anion states are known to be repulsive and sufficiently long lived to lead to dissociative electron attachment. They have been identified and located around 6.5 eV for 2B_1 , 8.6 eV for 2A_1 , and 11.8 eV for 2B_2 from the dissociative attachment cross-section measurements for the production of H^- (59–63), O^- (61–

63), and OH^- (61, 62). Of these, the B_1 state has been found to be the dominant resonant process in the total dissociative attachment cross section. Core-excited resonances at higher energy associated with Rydberg states arising from the promotion of a core electron into $3p$, $3d$, $4s$, . . . orbitals of the united atom have also been shown to occur in an energy range of 9–12 eV from electron transmission spectroscopy (64).

2. Multiple-site contribution

In the single-event approximation, where the various scattering amplitudes multiply reflected between different sites are neglected, the scattered intensity is given simply by the square of the sum of amplitudes that originate from the different sites. Overlapping of the scattering potentials (e.g. dipole term) yields a constant average potential in real space. Still, the elastic and inelastic contributions from each site differ from those of an isolated site because they include screening phenomena (e.g. induced polarization) due to the interaction of the site with its neighbors. Given that the electron is responsible for both the scattering potential, as a projectile, and the interaction among the sites, as an atomic or molecular component, the importance of the latter interaction may somehow be recognized from the short- or long-range character of the scattering potential. Within the single-event approximation, the electron wavelength criterion is a necessary and sufficient condition for the occurrence of interference effects in the elastic scattering, whereas in the inelastic case the formation of excitons is also necessary. Let us first look at the interference effects, then examine screening phenomena, and finally examine the consequences of the overlapping of the scattering potential more specifically in connection with resonant scattering.

The differential scattering cross section per scatterer for an incident particle of reduced mass m and wave-vector state $|\mathbf{k}_0\rangle$ to be scattered by a medium into a final state $|\mathbf{k}\rangle$, within a unit solid angle $d\Omega$ and unit energy interval dE , is given within the first Born approximation by (65, 66)

$$\frac{d^2\sigma}{d\Omega dE} = \frac{|\mathbf{k}|}{|\mathbf{k}_0|} \frac{m^2}{4\pi^2\hbar^5 N} |V(\mathbf{K})|^2 S(\mathbf{K}, \omega). \quad (13a)$$

In this equation, $\hbar\mathbf{K} = \hbar\mathbf{k}_0 - \hbar\mathbf{k}$ is the momentum transferred between the particle and the medium; $V(\mathbf{K})$ is the matrix element, between the initial and final state of the particle, of the potential describing the interaction of the particle with individual components of the medium; $\hbar\omega = E_0 - E$ is the energy transferred between the particle and the medium; N denotes the number of scatterers. Thus the cross section can be separated into a factor $\propto |V(\mathbf{K})|^2$, which depends only on the properties of the individual components of the medium, and a dynamic structure factor $S(\mathbf{K}, \omega)$, which depends on the geometrical arrangement as well as the collective dynamic properties of the medium. Considering more specifically $V(\mathbf{K})$ as reflecting elastic scattering from an atomic or molecular site, $S(\mathbf{K}, \omega)$ reduces

to depend only on the geometrical arrangement and dynamic properties of the atomic or molecular lattice. In this case, if multiple scattering is partially summed within a site, $V(\mathbf{K})$ can be replaced by the T -matrix transition amplitude (67). As such, this should more closely represent the behavior of the quasielastic cross sections in the single-event framework.

The dynamic structure factor is the space–time Fourier transform of the atomic or molecular time-dependent pair-distribution function, $G(\mathbf{x}, t)$, through the relationship (65, 66)

$$S(\mathbf{K}, \omega) = \frac{N}{2\pi} \int dt \exp(-i\omega t) \times \int d^3x \exp(i\mathbf{K} \cdot \mathbf{x}) G(\mathbf{x}, t). \quad (13b)$$

The instantaneous part of the pair distribution function, $G(\mathbf{x}, 0)$ (i.e. the static approximation), applies to the sum of elastic and quasielastic scattering of X rays in solids, liquids and gases (66, 68–70). In view of the disorder in an amorphous solid, the spatial range of $G(\mathbf{x}, 0)$ can be used to define an effective scattering volume beyond which quantum interference effects are negligible. For $t \neq 0$, the degree of correlation is generally expected to be reduced because the location of the atomic or molecular sites may change with time in a random fashion, as for instance in the case of a liquid. The elastic cross section, which is obtained by putting $\omega = 0$ in Eq. (13b) i.e. $S(\mathbf{K}, \omega = 0)$, corresponds to the time average of $G(\mathbf{x}, t)$. For a solid, this quantity still reflects the geometrical arrangement, but for the atoms or molecules moving about their equilibrium positions, as opposed to the frozen picture (i.e. snapshot) in the case of $t = 0$. On the other hand, $S(\mathbf{K}, \omega \neq 0)$, which is associated with various excitation modes of the nuclei, is normally less sensitive to the structure of the medium. This is best seen in the limit of uncorrelated vibrational excitation modes, such as weakly coupled local oscillators or localized modes induced by disorder. In this case, as the electron energy is transferred exclusively on a single site during scattering, the interference effects among different sites cannot arise. As a consequence, the cross section for the excitation of the various intramolecular vibrational modes in amorphous ice, especially those strongly coupled by hydrogen bonding but localized by disorder, should not strongly reflect the geometrical arrangement of the medium.

The modification brought about by the interaction among the atomic or molecular sites to the elastic and inelastic differential cross sections for electron scattering from an isolated site can be visualized simply with the following considerations. Regarding the interaction of the incident electron with the individual components of the medium in Eq. (13a) as given by the Coulomb interaction, $V(\mathbf{K}) = 4\pi e^2/K^2$, the dynamic structure factor, $S(\mathbf{K}, \omega)$, can be shown to be connected to the wave-number frequency-dependent dielectric constant, $\epsilon(\mathbf{K}, \omega)$, with (65)

$$\frac{4\pi^2 ne^2}{NK^2\hbar} [S(\mathbf{K}, -\omega) - S(\mathbf{K}, \omega)] = \text{Im} \left[\frac{1}{\varepsilon(\mathbf{K}, \omega)} \right] \\ = \frac{-\varepsilon_2(\mathbf{K}, \omega)}{|\varepsilon(\mathbf{K}, \omega)|^2} \quad (14)$$

In this expression, $\varepsilon(\mathbf{K}, \omega)$ is the dielectric constant, which includes all interactions among the components with $\varepsilon_2(\mathbf{K}, \omega)$ its imaginary part, and n is the number of atoms or molecules (i.e. N) per unit volume.

In the self-consistent field (SCF) approximation (71), $\varepsilon(\mathbf{K}, \omega)$ is given for an atomic or molecular solid simply by (72, 73)

$$\varepsilon(\mathbf{K}, \omega) = \lim_{\delta \rightarrow 0} 1 + \frac{4\pi ne^2}{\hbar K^2} \sum_f |\eta_{f0}(\mathbf{K})|^2 \\ \times \left[\frac{1}{(\omega_f - \omega_0) - \omega - i\delta} + \frac{1}{(\omega_f - \omega_0) + \omega + i\delta} \right], \quad (15a)$$

where $\eta_{f0}(\mathbf{K})$ is an atomic or molecular matrix element,

$$\eta_{f0}(\mathbf{K}) = \left\langle f \left| \sum_i \exp(i\mathbf{K} \cdot \mathbf{r}_i) \right| 0 \right\rangle, \quad (15b)$$

taken between an initial state taken here as the ground state $|0\rangle$ of energy ω_0 and an excited state $|f\rangle$ of energy $\hbar\omega_f$ with $\mathbf{r}_i$ being the position the i th atomic or molecular electron. This approximation amounts to a neglect of local-field corrections by taking the field at each site to be the averaged macroscopic field. In other words, all long-range SCF effects are included, whereas the short-range screened interactions between electrons are smoothed out.

Using further the relationship $(\omega - \omega_{f0} + i\delta)^{-1} = P(\omega - \omega_{f0})^{-1} - i\pi\delta(\omega - \omega_{f0})$ with $\lim \delta \rightarrow 0$, where P represents the Cauchy principal value and $\omega_{f0} = \omega_f - \omega_0$, to separate the real and imaginary part of $\varepsilon(\mathbf{K}, \omega)$ in Eq. (15a), and substituting the latter in Eq. (14), we obtain the following expression for the dynamic structure factor:

$$S(\mathbf{K}, \omega) = \frac{N}{|\varepsilon(\mathbf{K}, \omega)|^2} \sum_f |\eta_{f0}(\mathbf{K})|^2 \delta(\omega - \omega_{f0}) \\ = \frac{N}{|\varepsilon(\mathbf{K}, \omega)|^2} \hbar |\eta_{E_0-E}(\mathbf{K})|^2 \quad (16)$$

In this last step, we have introduced with $E_0 - E = \hbar\omega$ the quantity $|\eta_{E_0-E}(\mathbf{K})|^2$, which is the form factor for a single atom or molecule (74). Inserting the latter expression for $S(\mathbf{K}, \omega)$ in Eq. (13a) along with the Coulomb interaction, $V(\mathbf{K}) = 4\pi e^2/K^2$, we finally obtain, for differential scattering cross section per scatterer in the condensed phase,

$$\frac{d^2\sigma}{d\Omega dE} = \frac{|\mathbf{k}|}{|\mathbf{k}_0|} \frac{m^2}{4\pi^2\hbar^4} |V(\mathbf{K})|^2 \times |\eta_{E_0-E}(\mathbf{K})|^2 \\ \times \frac{1}{|\varepsilon(\mathbf{K}, \omega)|^2} \quad (17)$$

In this expression, the first part corresponds to the Rutherford formula, which applies to the scattering of an electron with a free electron being at rest. The second part $|\eta_{E_0-E}(\mathbf{K})|^2$, which has been introduced above, taken along with the first factor can immediately be recognized as the differential cross section for electron scattering on a single atom or molecule (i.e. gas phase). Finally, the third part, referred to as a screening factor, can modify the cross section given by the two preceding factors. Depending upon the value taken by the real $\varepsilon_1(\mathbf{K}, \omega)$ and imaginary part $\varepsilon_2(\mathbf{K}, \omega)$ in this factor, one may distinguish two different cases: a screening (i.e. protecting) effect when $\varepsilon_1^2 + \varepsilon_2^2 > 1$, or an antiscreening (i.e. sensitizing) effect when $\varepsilon_1^2 + \varepsilon_2^2 < 1$. It is worth mentioning here, since this equation was derived with only the electronic excitations in mind [i.e. $V(\mathbf{K}) = 4\pi e^2/K^2$], that the same equation applies in principle to vibrational excitations with the appropriate electron-molecule interaction potential $V(\mathbf{K})$ and vibrational matrix elements $\eta_{f0}(\mathbf{K})$.

Owing to the overlapping of the electron scattering potentials in the condensed phase (e.g. dipole term), an electron resonance considered as a scattering event localized on a site is also perturbed since it is coupled to various longitudinal polarization modes of the surrounding medium. In other words, this coupling can be seen as the scattering of a resonant electron on a site by the long-range part of the interaction potential about another site. Since this latter effect implies a scattering amplitude reflected between two sites, it occurs at a higher order of scattering than in the single-event approximation. On the other hand, from the point of view of the polarization modes, the resonance may simply be seen as a time-dependent electrostatic perturbation, which results in virtual (i.e. resonance-energy shift) (75) and real excitations (i.e. resonance-energy transfer) (76, 77) of the surrounding medium. For a classical harmonic oscillator of angular frequency ω_i driven by a time-dependent force $F(t)$, the energy transferred to the oscillator is given by (76)

$$E_0 - E = \frac{1}{2M_r} \left| \int_{-\infty}^{+\infty} F(t) \exp(i\omega_i t) dt \right|^2, \quad (18)$$

where M_r is the reduced mass of the oscillator.

Quantum mechanically, Eq. (18) can be interpreted as the average energy $\langle E_0 - E \rangle$ transferred to an oscillator, with $\langle E_0 - E \rangle / \hbar\omega_i$ corresponding to the excitation probability P_n for an oscillator ω_i initially in its ground state $|0\rangle$ to be found in the state $|n\rangle$ after the resonance decay. Two regions can be delineated from Eq. (18). For a resonance lifetime short *relative* to the period of the oscillator, only a small amount of energy is transferred to the excitation

mode, and it results in a small P_n . Conversely, if the resonance lifetime is long compared to the period of the oscillator and the electrostatic interaction is sufficiently large, reasonable energy transfer occurs and P_n increases accordingly. In other words, if the contribution to the polarization energy $\langle E_0 - E \rangle$ brought about by a longitudinal polarization mode is much smaller or greater than its energy $\hbar\omega_n$, the mode is classified as contributing to either the shift or the broadening of a resonance, respectively.

Total Cross Section

The total cross section for electron scattering on water in the gas phase (25–27), shown as open symbols in Fig. 3, is found to rise sharply below 3 eV and to exhibit a broad maximum around 9 eV. The present results, which compare in size to those of Kurtz *et al.* in ice (24), are in general about one order of magnitude smaller than the gas-phase data. However, the comparison between the gas and ice data cannot be made too directly, because the various nuclear and electronic excitation modes of a water molecule are coupled in the condensed phase. For instance, owing to the strong hydrogen bonding in ice, the rotational and translational degree of freedom of the molecules are hindered, the intramolecular-vibrational modes are strongly perturbed, all sharp electronic excitations seen in the gas phase disappear, and finally the lowest excited state are blue-shifted by 1.1–1.5 eV. Furthermore, owing to the quantum-mechanical nature of the electron, the cross sections for low-energy electron scattering in ice must intrinsically include correlation effects related to the proximity of the molecular scattering sites. Keeping these notions in mind, the difference in the magnitude of the cross section between the gas and condensed phase is discussed separately for the elastic, vibrational and electronic contributions in the following section.

Integral Scattering Cross Sections

1. Elastic and phonon cross sections

The elastic cross section illustrated in Fig. 5a corresponds to the angularly integrated isotropic component, $2Q_{re}$. It reveals a rise at low energy and three broad structures whose maxima are located at about $E_0 = 6.5$, 14.5 and 40 eV, along with a small shoulder around 80 eV. Except for the oscillatory structure, the present cross section is up to about two orders of magnitude smaller than the measured integral elastic cross sections in the gas phase, which are shown as open symbols for two different sets (53, 55) in Fig. 5a. However, it must be noted that the latter includes translational and rotational energy losses. This is not the case here, because the elastic peak is well resolved from the various phonon energy losses. Furthermore, as mentioned in the section on the multiple-scattering model, forward elastic scattering, which is known to be a strong component in the gas phase (53, 55), is not contributing to

the electron-beam attenuation and therefore to the present elastic cross section.

The translational phonon modes are taken as a narrow energy distribution at about $\hbar\nu'_T = 10$ meV and of FWHM of 1 meV along with another centered around $\hbar\nu''_T = 24$ meV and of FWHM of 25 meV. The $\hbar\nu'_T$ mode, which has a relatively small cross section, contributed to improving the fit in Fig. 4 and will not be discussed further. The energy dependence of the cross section for the $\hbar\nu''_T$ mode shown in Fig. 5a exhibits a strong rise at low energy along with an oscillatory structure similar to that of the elastic cross section. The value of the corresponding coefficient of angular anisotropy γ (Table 2) presents a slow but steady increase up to about 100 eV. Hence the excitation of phonons by the electron via the permanent dipole of the molecule should be important over the present energy range and especially at low incident energy, where a rise in the cross section occurs.

The librational phonon modes are described by two energy distributions at about $\hbar\nu'_L = 61$ meV of FWHM of 30 meV and $\hbar\nu''_L = 92$ meV of FWHM of 40 meV. Both cross sections also exhibit a rise at low energy. This is followed by a maximum around 7.0 eV in the case of the ν'_L mode, which is similar to the ν''_T mode, and a broader maximum around 8.5 eV for the ν'_L mode. This difference in the energy of the maximum allows us to resolve them in the energy-loss spectra at an incident energy of 9 eV (13). At higher incident energy, both libration cross sections show essentially the same oscillatory structure, which is also reminiscent of the elastic scattering cross section. The corresponding coefficients of angular anisotropy γ (Table 2) increase with the incident energy more quickly than that for the translational phonon mode and reach nearly pure forward scattering conditions at 25 eV. This indicates that the direct electron scattering process via the permanent dipole of the molecule should contribute significantly to the excitation of the libration modes above 25 eV.

Between 1.5 and 100 eV, the electron MFP (Table 1) in amorphous ice is typically 4 to 10 times larger than both the de Broglie wavelength of the electron and the distance between the scatterers. As such this result must be considered a necessary condition to justify *a posteriori* the single-event approach for the calculation of the scattered intensity. Also the modulus of the maximum wave-vector transfer $|\mathbf{K}_{\max}| = 2|\mathbf{k}_0|$ varies between $1 \text{ \AA}^{-1}$ and $10 \text{ \AA}^{-1}$. Within the same range, the static structure factor $S(\mathbf{K})$ of amorphous ice at 10 K (20) exhibits several oscillatory structures whose maxima are located at $|\mathbf{K}|_1 = 2 \text{ \AA}^{-1}$, $|\mathbf{K}|_2 = 3.1 \text{ \AA}^{-1}$, $|\mathbf{K}|_3 = 5 \text{ \AA}^{-1}$, and $|\mathbf{K}|_4 = 7.5 \text{ \AA}^{-1}$. This suggests that some energy features common among different cross sections in Fig. 5a could originate from interference effects and as such be connected to the static structure factor of amorphous ice. How the static structure factor can be embodied into the present cross sections may be understood simply with the following considerations.

In the case of an electron being scattered quasielastically from a condensed medium, Eq. (13a) reduces to

$$\frac{d\sigma_{qe}}{d\Omega} = \frac{m^2}{4\pi^2\hbar^4N} |V(\mathbf{K})|^2 \int_{\Delta\omega} S(\mathbf{K}, \omega) d\omega, \quad (19)$$

upon assuming that $|\mathbf{k}_0| = |\mathbf{k}|$. Performing the integration over a small energy transfer range $\Delta\hbar\omega$, so as to include the elastic and all the phonon modes, the integral leads to the following two terms, $N\delta_{\mathbf{k},0} + S(\mathbf{K})$. The static structure factor $S(\mathbf{K})$, which accounts for the quantum interferences among the scattering sites, is defined upon subtracting the $N\delta_{\mathbf{k},0}$ term. The latter term corresponds to the square of the sum of amplitudes that interfere constructively only in the forward directions, $\mathbf{K} = 0$. Also, since this term amounts to a constant average potential that is not contributing to scattering within the medium, it will not be considered further. Alternatively, from the space–time Fourier relationship (Eq. 13b), it can be seen that $S(\mathbf{K})$ is also connected to $G(\mathbf{x}, 0)$. For an amorphous solid, where the individual scattering sites are orientationally disordered, some simplifications are in order, and only the modulus of the wavevector transfer, $K = |\mathbf{k}_0|\sqrt{2(1 - \cos\theta)}$, instead of $\mathbf{K}$, is necessary. Here θ is the angle between the incident $\mathbf{k}_0$ and scattered $\mathbf{k}$ electron wavevectors. Hence for each K , $S(K)$ keeps the same value along the continuous curve $K = |\mathbf{k}_0|\sqrt{2(1 - \cos\theta)}$ in the $|\mathbf{k}_0|$, θ plane and similarly in the E_0 , θ plane with $|\mathbf{k}_0|^2 \propto E_0$. Moreover, in the presence of maxima and minima in $S(K)$ and for a fixed incident energy E_0 , the value of $S(K) = S(E_0, \theta)$ exhibits several angular lobes as a function of θ . An angular lobe at about a scattering angle θ occurs where the line E_0 intersects the continuous curve corresponding to a maximum in the E_0 , θ plane. With increasing E_0 , the angular lobes converge progressively toward smaller scattering angles along with a new one emerging at large scattering angle. Integrating the differential cross section Eq. (19) further over the whole angular range, we get the integral quasielastic cross section per scatterer,

$$\sigma_{qe}(E_0) = \frac{m^2}{4\pi^2\hbar^4} \int_0^{2\pi} \int_0^\pi |V(K)|^2 S(K) \sin\theta d\theta d\phi, \quad (20)$$

where both factors $V(K)$ and $S(K)$ depend on the incident energy E_0 and θ through K , but are independent of ϕ . While $V(K)$ relates to the electron scattering from a site, $S(K)$ accounts for the quantum interferences arising among the scattering sites.

The measured integral cross section corresponding to $\sigma_{qe}(E_0)$ for quasielastic scattering of electron in amorphous ice is shown in Fig. 7a. It is obtained from the sum of the elastic cross section and translational phonon contributions ν'_{τ} and ν''_{τ} in Fig. 5. The librational modes are not included here since the values of the static structure factor have been obtained from X-ray diffraction measurement, which is known to reflect mainly the geometrical arrangement of the oxygen sublattice (68).

Although, both $V(K)$ and $S(K)$ in Eq. (20) play a role in the behavior of $\sigma_{qe}(E_0)$, the importance of $S(K)$ can be grasped better if we consider the following scenario. Suppose that the contribution from a site is isotropic and the

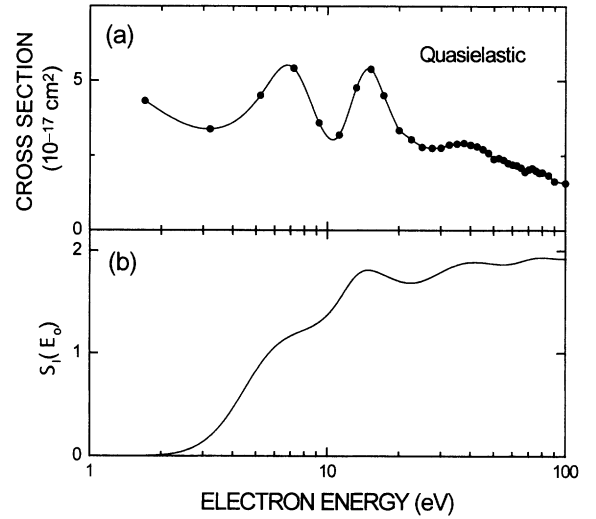


FIG. 7. Energy dependence of (panel a) the quasielastic integral cross section per scatterer, given by the sum of the elastic and translational phonon contributions ν'_{τ} and ν''_{τ} , for an electron propagating in amorphous ice, and (panel b) the integral factor $S_I(E_0)$, as calculated from the static structure factor $S(K)$ data of the low-density amorphous ice at 10 K, modulating the quasielastic integral cross section corresponding to an isolated site.

same as that for an isolated site (i.e. gas phase) and that the scattering potentials from different sites are not overlapping. This is like assuming that the range of the electron scattering potential from an isolated site is much smaller than the de Broglie wavelength of the electron and the closest distance separating the scattering sites, so that $V(K)$ in Eq. (20) can be replaced by $V(E_0)$, which is independent of θ . Integrating over ϕ in Eq. (20), the cross section per scatterer can immediately be written as

$$\sigma_{qe}(E_0) = \frac{m^2}{4\pi^2\hbar^4} |V(E_0)|^2 2\pi S_I(E_0), \quad (21a)$$

with

$$S_I(E_0) = \int_0^\pi S(E_0, \theta) \sin\theta d\theta \quad (21b)$$

standing for the θ angular integration as function of the incident energy.

The integral factor $S_I(E_0)$, which is the key quantity here, appears simply as a factor modulating the cross section corresponding to the gas phase for short-range scattering conditions. Its energy dependence is shown between 1 and 100 eV in Fig. 7b, as calculated from $S(K)$ data for amorphous ice at 10 K. The only adjustable parameter was the effective mass of the electron as it propagates in amorphous ice, m^* , which was fixed at $0.8m$ throughout the entire energy range. The factor $S_I(E_0)$ increases from zero at low energy to an asymptotic value of two at high energy. The latter limit, which is practically reached at 100 eV, amounts to the neglect of interference effects in the scattering from multiple sites [i.e. $S(K) = 1$]. At this energy, the cross section per

scatterer $\sigma_{qe}(E_0)$ turns out to be the same as it would be in the gas phase for short-range scattering conditions. Below 100 eV, $S_f(E_0)$ can be significantly smaller than two with the calculated $\sigma_{qe}(E_0)$ being smaller than its gas-phase counterpart. The oscillatory behavior results merely from the shifting of the angular distribution $S(E_0, \theta)$ with the energy E_0 combined to the weighting factor $\sin \theta$ in Eq. (21b). The behavior of $S_f(E_0)$ at low energy is most interesting. Notwithstanding the genuine angular contribution from an isolated site, which is isotropic in the present case, the electron propagation may acquire an anisotropic character with the development of scattering angular lobes. Better yet, owing essentially to its wavelike nature, an electron at low energy can propagate quasielastically in an amorphous solid more freely than one would expect from a purely classical basis.

The different behavior between the measured quasielastic integral cross section and $S_f(E_0)$ in Fig. 7 is attributed to the factor $|V(K)|^2$ in Eq. (20) (or more exactly its associated T -matrix), which generally depends on both the incident energy E_0 and θ . Also, for a longer-range scattering potential, $V(K)$, a larger difference is found between the calculated $\sigma_{qe}(E_0)$ and its gas-phase counterpart [i.e. $S(K) = 1$] (M. Michaud and L. Sanche, unpublished results). Such a result is expected in this case simply because the scattering angular distribution of $|V(K)|^2$ is predominant in the forward direction where the integration weight of $S(E_0, \theta)$ in Eq. (20) is negligible. Regardless of the latter interference effect, the magnitude of the quasielastic cross section in the medium is also expected to be smaller than its gas-phase analogue owing to screening from the librational, vibrational and electronic polarization modes of the surrounding medium.

2. Vibrational cross sections

Owing to the small energy separation between the symmetrical ν_1 and asymmetrical ν_3 stretching region, a single Gaussian distribution at about 417 meV and of 50 meV FWHM was used to represent both vibrational modes at all energies (Table 3). It was only necessary to shift this distribution slightly toward higher energy loss around 8 eV incident energy, where a noticeable enhancement occurs in the cross section as shown in Fig. 5b. Despite the smaller cross-section values, the same enhancement was found in the excitation function of the overtone $2\nu_{1,3}$. This enhancement at 8 eV can be explained by the presence of the broad resonance associated with the 2B_2 anion state of the molecule and whose maximum is at lower energy in the gas phase (i.e. 6–8 eV). Two distinct properties of the condensed phase of water can account for the different energy and lifetime of this resonance compared to the gas phase: the presence of hydrogen bonding and the electronic polarization of the surrounding medium. The effect of hydrogen bonding, which induces modifications in the geometry of a molecule in its ground state, may consist of a shift toward higher or lower energy of an electron resonance,

depending on the circumstances, along with a corresponding change in its lifetime. The electronic polarization usually causes energy lowering and broadening of a resonance through inhomogeneous distribution of polarization energies. The coefficient of angular anisotropy γ corresponding to the stretching modes (Table 3) increases steadily, from low to high incident energy, but is still smaller than that of the intermolecular vibrational modes. This indicates that a contribution to the cross section due to relatively long-range interaction potentials remains. The latter is expected to originate particularly from the dipolar term and might explain the small rise below 3 eV.

The bending mode ν_2 was represented by a 16 meV FWHM Gaussian distribution at about 204 meV (Table 3). The behavior of the ν_2 cross section shown in Fig. 5b is similar to that of the stretching modes with the vibrational enhancement at 2B_2 resonance position but does not rise at low energy. The coefficient of angular anisotropy γ (Table 3) follows the same general trend as the stretching modes but is 1.5 greater on average, thus indicating a somewhat larger contribution due to the long-range interaction potential. These results are consistent with gas-phase data, where resonance enhancement is found to be more pronounced in the stretching than in the bending-mode cross section (50, 57) and the dipole interaction makes a larger contribution.

Below 20 eV, the present stretching modes $\nu_{1,3}$ and bending mode ν_2 cross sections (Table 3) amount to about 25% of the measured integral cross sections in the gas phase (50, 52) shown as open symbols in Fig. 5b. For energies higher than 20 eV, the bending cross section becomes larger than the stretching one, as calculated in the gas phase (56). At 50 eV, the former reaches about 40% of the predicted gas-phase value while the latter becomes comparable (56). Apart from possible interference effects, the magnitude of the vibrational scattering cross sections is expected to be generally reduced in the condensed phase, owing to the screening from the vibrational as well as electronic polarization modes of the surrounding medium. Considering only the latter contribution to be effective over the vibrational energy transfer range, with an optical dielectric constant of $\epsilon_1(0, \omega) \cong 1.8$ and $\epsilon_2(0, \omega) = 0$ as obtained for liquid water (78), one readily finds a value $3 < \epsilon_1^2 + \epsilon_2^2 < 4$ for the attenuation factor in Eq. (17). However, in view of the wave-number dependence of the dielectric constant $\epsilon(\mathbf{K}, \omega)$, the latter figure might be somewhat different with the contribution from large momentum transfer (i.e. the short-range screened interactions). Besides, as a corollary, considering that the electron–molecule interaction potential $V(\mathbf{K})$ and the vibrational matrix element $\eta_{j0}(\mathbf{K})$ may have either a short- or long-range character, electron-impact excitation of distinct vibrational modes should be screened differently.

3. Electronic cross sections

The sum of the scattering cross sections attributed to all other inelastic features existing outside the present vibra-

tional spectra is shown in Fig. 6. This cross section exhibits a small hump around 5.2 eV followed by a steady rise above the electronic excitation threshold at 7.5 eV. It already accounts for 40% of the total inelastic cross section at 20 eV and 70% at 100 eV (Table 3). Since there is no neutral excited electronic state of water below 6 eV in both gas (79) and condensed phase (80), the small hump can only be ascribed to the formation of the 2B_1 transient anion state. This state is known to be repulsive and sufficiently long lived to lead to the production of H^- upon low-energy electron impact on the water molecule (59, 60) as well as on multilayer films of ice (81). Along with the dissociative attachment process, this state is also expected to decay into high vibrational levels of the ground state at energies lying outside the present energy-loss range (i.e. 1 eV). The steady rise in the cross section above 7.5 eV is attributed to various electronic excitations and ionization processes. The electronic excitations involve mostly valence-type transitions such as $^3,1B_1(1b_1 \rightarrow 3s4a_1)$ at 8.5 eV, $^3,1A_1(3a_1 \rightarrow 3s4a_1)$ along with the $^3,1A_2(1b_1 \rightarrow 3p2b_2)$ around 10.4 eV, $^3,1B_2(1b_2 \rightarrow 3s4a_1)$ at 14.5 eV, and the $^3,1A_1(2a_1 \rightarrow 3s4a_1)$ ~ 28 eV (80, 82, 83). It should be noted that the large energy absorption band found at ~ 20 eV in the optical and electron energy-loss spectra of ice, often attributed to excitation of plasmons, has been shown theoretically (84) and more recently in an experimental study (85) to depart from the expected plasmon behavior. From the spectrum of secondary electrons in the latter work, autoionization instead of plasmon decay was suggested as the major energy loss process.

Experimental information on the electron scattering cross section for electronic excitations in the gas phase is scarce. A set of data, which has been reported from indirect methods, is divided into four components corresponding somewhat to the above excitations (86). Each component exhibits, slightly above its electronic excitation threshold, a maximum, having values of $\sim 1.4 \times 10^{-16} \text{ cm}^2$, $\sim 0.1 \times 10^{-16} \text{ cm}^2$, $\sim 0.1 \times 10^{-16} \text{ cm}^2$, and $\sim 0.7 \times 10^{-16} \text{ cm}^2$, shown respectively as crosses in Fig. 6. The ionization processes can be ascribed to the three outermost molecular orbitals, $1b_1$, $3a_1$ and $1b_2$, having lower binding energies in the condensed phase at 10.9, 13.5 and 17 eV, respectively (87), along with a contribution of the $2a_1$ orbital located at 32.6 eV in the water vapor (88). The total electron impact ionization cross section measured in the gas phase, which is shown for two different sets of data (89, 90) over the present energy range in Fig. 6, shows a rise from 10^{-17} cm^2 at about 15 eV to a maximum of $2 \times 10^{-16} \text{ cm}^2$ at 100 eV. A single ionization process should contribute to about half of this cross section (91).

Within the incident energy range from 7.5 to 20 eV, the present inelastic scattering cross section corresponds to roughly 10% to 70% of the summed electronic excitation and ionization cross section, as reported in the gas phase (86). This much lower value concurs with the disappearance of all the sharp electronic excitations and the screening

from the electronic polarization modes of the surrounding medium. Considering only the latter effect with the value of the optical dielectric constant for liquid water for the energy transfer ranging between 7.5 and 20 eV (78), one readily finds from Eq. (17) the value $1.3 < \epsilon_1^2 + \epsilon_2^2 < 7$ for the screening factor. Therefore all energy-loss cross sections having their incident energy threshold lying within this energy range are under the effect of a strong screening. At higher incident energy between 20 and 25 eV, our measurement reaches approximately 80% of the ionization cross section measured in the gas phase (89) at about $5 \times 10^{-17} \text{ cm}^2$. Also, within the energy transfer ranging from 20 to 25 eV, the screening factor exhibits somewhat of an "anti-screening" behavior with a value $0.5 < \epsilon_1^2 + \epsilon_2^2 < 1$. As a result, the energy-loss cross sections characterized by their incident energy thresholds lying between 20 and 25 eV end up being closer to their gas-phase counterparts. Above 30 eV, our measurements, which suffer from both screening and anti-screening effects, averages about 50% of the gas-phase values. At higher incident energy, one would have to account for the effect of the allowed larger momentum as well as energy transfers in $\epsilon(\mathbf{K}, \omega)$.

CONCLUSIONS

We have shown that, under reasonable experimental conditions, it is possible to extract elastic and inelastic absolute cross sections per scatterer or scattering probabilities per unit length for 1–100 eV electron scattering in amorphous ice by using a two-stream multiple scattering analysis of HREEL spectra of vacuum deposited films. Assessment is made for incoherent elastic and inelastic electron scattering as well as for the superimposition of broad energy-loss features. The angular distribution of a scattering event is also included by means of a coefficient of angular anisotropy that can simulate both the short- and long-range interaction effectively (e.g. dipole scattering). The cross sections so extracted include intrinsically coherent scattering from a single site along with interference and screening effects originating from the presence of neighboring sites. The former effect is found to result into resonant scattering similarly to the gas phase. The presence of neighboring sites is described, within the single-event approach for the calculation of the scattering probabilities, by the dynamic structure factor $S(\mathbf{K}, \omega)$. This is expected to be a valid approach when the electron MFP is much larger than both the de Broglie wavelength of the electron and the distance between the scatterers. Judging from the values extracted for the electron MFP in amorphous ice between 1.5 and 100 eV (Table 1), which compare in magnitude to those in ice of Kurtz *et al.* (24), this condition seems to be reasonably fulfilled in the present study.

The oscillatory feature found in the cross section for quasielastic (i.e. elastic plus phonon) scattering of electrons in amorphous ice is attributed essentially to the wavelike nature of the propagating electron, which produces quan-

tum interference effects. We show more specifically, with a simple model, that the oscillatory feature reflects the behavior of the static structure factor $S(K)$ of amorphous ice with respect to incident electron energy. Furthermore, the same interference effects can partly elucidate two legitimate remarks that one might have about the present experimental quasielastic cross section: the much smaller value obtained for the cross section for quasielastic scattering at very low energy range (i.e. <10 eV) in the condensed phase compared to that in the gas phase, and the relatively large coefficient of angular anisotropy found in going at higher incident energy.

The dynamic structure factor, $S(\mathbf{K}, \omega)$, when connected to the wave number and frequency-dependent dielectric constant, $\epsilon(\mathbf{K}, \omega)$, can further be seen to be responsible for the screening of the interaction of an electron with an atom or molecule by its neighbors in the condensed phase. Therefore, regardless of the interference effect, $\epsilon(\mathbf{K}, \omega)$ is shown to have an immediate effect on the magnitude of the elastic and inelastic electron scattering cross sections that would originate from non-interacting sites (i.e. isolated targets). The magnitude of the present elastic and inelastic cross sections are generally found to be smaller than their gas-phase analogues, owing essentially to the prevalent screening (i.e. $\epsilon_1^2 + \epsilon_2^2 > 1$) of the electron-atom or -molecule interaction potential in the condensed phase.

The present work concerns mostly the interaction of secondary low-energy electrons with solid water (i.e. amorphous ice) at very low temperature along with a comparison directly to the gas phase (i.e. isolated molecule). However, it may still be possible to infer from the present study what to expect in terms of similarities and differences between the amorphous ice and liquid water. In amorphous ice, the molecules do not remain rigidly fixed in space, as they possess some kinetic energy, which increases with the temperature. At very low temperature, this thermal energy is not sufficient to enable the molecules to go very far from their ideal equilibrium positions. The main effect of this energy is therefore to allow the molecule to undergo only small translational and librational vibrations about these positions. As the temperature rises, the amplitude of these vibrations increases, and eventually at the melting temperature the alteration of the molecular arrangements from the ideal molecular lattice becomes the most important source of the temperature dependence of electron-scattering mechanisms. Liquid water at room temperature exhibits an X-ray static structure factor $S(K)$ along with a molecular arrangement around central oxygen quite similar to that of amorphous ice (20). It appears that the thermal fluctuations in the liquid phase would be taking place over distances large enough so that the typical pair-correlation distances among the molecules would still be reasonably preserved. Besides, the optical spectra of liquid water exhibit phonon bands at nominal positions of 60 and 170 cm^{-1} along with librational bands near 450, 550 and 720–740 cm^{-1} (92), quite similar to those in amorphous ice. So, in a sense,

liquid water might merely be seen as thermally agitated amorphous ice, but where the molecules could possibly hop from one lattice site to another.

Considering the similar molecular layout in both phases, the transport part of the electron propagation in the liquid phase can also be described through a classical approach. However, owing to the effect of the temperature, one must account for the stochastic element of scattering probabilities per unit length or cross sections for momentum energy given as well as *absorbed* by the electron. Likewise, the scattering cross section can be separated into a factor $\propto |V(\mathbf{K})|^2$, which depends only on the properties of the individual components of the medium, and a dynamic structure factor, $S(\mathbf{K}, \omega)$, which regards more specifically the collective properties (e.g. translational, vibrational and electronic excitations) of the liquid. The difference between electron scattering in the solid and liquid phase is expected to be due mainly to the intermolecular (i.e. translational and librational) vibrational excitations, which are thermally excited at room temperature. On the other hand, electron scattering due to intramolecular vibrational excitations, electronic excitations, along with the electronic screening phenomena in liquid water, which depends mainly upon changes in the molecular density, should be somewhat the same as in amorphous ice.

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